

Ibne Farabi Shihab, Ph.D.

Applied Scientist II, Amazon
AI · Computer Vision · Transportation Systems

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RESEARCH OVERVIEW

My research advances artificial intelligence at the intersection of computer vision, Large Language Models, and safety-critical transportation systems. I develop interpretable and efficient AI methods that integrate multimodal learning, reinforcement learning, and quantum computing to enhance real-time decision-making in autonomous systems. My work addresses fundamental challenges in video understanding, crash detection, and next-generation network security, with practical applications demonstrated through industry partnerships and federal funding.

EDUCATION

- **Ph.D., Computer Science, Ames, IA, Iowa State University** October 2025
Dissertation: "Resource-Efficient AI for Real-World Applications: Model Compression, Sample-Efficient Learning, and Inference Optimization Across Traffic Safety, Quantum Systems, and Sequential Decision-Making"
Advisor: Dr. Anuj Sharma
- **M.S., Artificial Intelligence, Ames, IA, Iowa State University** January 2024
Thesis: "Robust and precise sidewalk detection with ensemble learning: Enhancing road safety and facilitating curb space management"
Advisor: Dr. Anuj Sharma
- **B.Sc., Computer Science & Engineering, Dhaka, Bangladesh, BRAC University** August 2018
Thesis: "Where will you setup your business next?: a machine learning approach to suggest ideal geographical location for new restaurant establishment"
Advisor: Dr. Hossain Arif

RESEARCH INTERESTS

- **Efficient & Reliable LLMs:** Structured pruning of state-space models, subspace-aware compression, and PAC-Bayes risk certificates.
- **Causal & Verifiable Reasoning:** Causal consistency regularization, constraint propagation, and detecting reward hacking in LLM alignment.
- **Video Understanding for Traffic Safety:** Sub-second temporal localization of crash events and LLM-guided crash narrative generation.
- **Reinforcement Learning:** Sparse-reward tractability, complexity-aware regularization, and LLM-guided policy optimization.
- **Trustworthy ITS & Calibrated Perception:** Conformal prediction, calibrated super-resolution, and robust ensemble perception for driver and roadway analysis.
- **Quantum ML & Network Security:** Few-shot learning for quantum spin systems and quantum neural networks for zero-trust 7G architectures.

RESEARCH EXPERIENCE

- **Graduate Research Assistant, Iowa State University, Ames, IA** January 2022 – Oct 2025
 - **Transportation Safety Research:** Led DOT-funded projects with Iowa DOT (\$1M+ total funding) developing AI-based navigation systems for snowplow operations, real-time crash detection frameworks, and ADAS enhancement under extreme weather conditions. First-/co-authored papers at IEEE ITSC ('23, '25 ×2), CVPR Workshops '24, IEEE T-ITS '26, and TRC-30 '24.
 - **Computer Vision & LLM Integration:** Pioneered novel frameworks integrating computer vision with Large Language Models for automated crash narrative generation, document layout analysis, and structured reasoning. First-authored papers at **ACL 2026 (3 Main + 1 Findings)**, **CVPR 2026**, **EMNLP 2025 (2)**, and **IEEE T-ITS 2026**. Developed synthetic crash video simulation pipeline for enhanced model training.

- **Quantum Computing Applications:** Developed innovative Quantum Neural Network architectures and self-supervised few-shot learning methods for quantum spin systems. First-authored **ECAI 2025** paper; co-authored work in *Measurement: Digitalization* (2025) on quantum-driven zero-trust architectures, increasing anomaly detection accuracy by 10% for next-generation network security.
- **Interdisciplinary Research:** Implemented reinforcement learning algorithms for cellular simulations optimizing CAR T-cell therapy parameters, published in *Biotechnology and Bioengineering* (2024, IF: 3.8) and *Biochemical Engineering Journal* (2022).

TEACHING EXPERIENCE

- **Lecturer, Dallas College, Dallas, TX** *August 2025 – January 2026*
 - **Course Instruction:** Taught Programming Fundamentals in C++, Java, and Kotlin (COSC 1436, COSC 1437, ITSE 1329) across multiple sections. Developed project-based curriculum integrating real-world applied AI examples.
 - **Student Mentorship:** Maintained 8 hours/week office hours; mentored students on capstone and independent projects.
- **Graduate Teaching Assistant, Iowa State University, Ames, IA** *August 2020 – December 2021; September 2024 – May 2025*
 - **Advanced Technical Courses:** Served as TA for Motion Planning for Robotics and Autonomous Systems (COM S 576), Advanced Programming Techniques in C++ (COM S 327), and Introduction to Programming in Java (COM S 113). Designed hands-on labs in ROS/CARLA; created PyTorch assignments on Video-LLMs.
 - **Course Development:** Contributed to curriculum development for Machine Learning and Deep Learning courses, designing programming assignments and assessment materials. Created guest lecture on RL for Snowplow Navigation (1.2K+ YouTube views).
 - **Student Mentorship:** Supervised 5 MS students on research projects; 2 co-authored publications at ITSC.
- **Graduate Teaching Assistant, University of Vermont, Burlington, VT** *August 2019 – May 2020*
 - **AI Education:** Delivered comprehensive instruction in machine learning and deep learning fundamentals, focusing on mathematical foundations and practical implementations.

PUBLICATIONS

*denotes equal contribution. All works below were produced during my Iowa State Graduate Research Assistantship (Jan 2022 – Oct 2025) under Dr. Anuj Sharma.

Published & Accepted

- **Shihab, I.F.**, Akter, S., Bhagat, S., Wood, J., and Sharma, A. “Beyond Accuracy: Empirical Spectral Diagnostics for Crash Classification Models.” **IEEE ITSC 2026**. arXiv:2602.19528
- Akter, S., **Shihab, I.F.**, and Sharma, A. “LLM-Guided Meta-Prior Reinforcement Learning for Adaptive Traffic Signal Synchronization.” **IEEE ITSC 2026**.
- Bhagat, S., Kandiboina, R., **Shihab, I.F.**, Knickerbocker, S., Hawkins, N., and Sharma, A. “Improving crash data quality by identifying misclassified alcohol-involved crashes using NLP on narrative data, **Journal of Safety Research**, Volume 97, 2026, Pages 777-791, ISSN 0022-4375,” <https://doi.org/10.1016/j.jsr.2026.05.007>
- **Shihab, I.F.**, Akter, S., and Sharma, A. “CalPro: Prior-Aware Evidential Conformal Prediction with Structure-Aware Sensitivity Bounds for Protein Structures.” **ICML 2026** .arXiv:2601.07201
- **Shihab, I.F.**, Akter, S., and Sharma, A. “What Reward Structure Enables Efficient Sparse-Reward RL? A Proof-of-Concept with Policy-Aware Matrix Completion.” **ICML 2026**. arXiv:2509.03790
- **Shihab, I.F.**, Akter, S., and Sharma, A. “CGRiC: Compositional Risk Certification for Structured LLM Outputs.” **ICML 2026**.
- **Shihab, I.F.**, Akter, S., and Sharma, A. “Certificate-Guided Pruning for Stochastic Lipschitz Optimization.” **ICML 2026**. arXiv:2601.20231
- **Shihab, I.F.***, Akter, S.*, and Sharma, A. “Adaptive Constraint Propagation: Scaling Structured Inference for Large Language Models via Meta-Reinforcement Learning.” **ACL 2026** (Main Conference). arXiv:2601.00095
- **Shihab, I.F.***, Akter, S.*, and Sharma, A. “Beyond Variance: Knowledge-Aware LLM Compression via Fisher-Aligned Subspace Diagnostics.” **ACL 2026** (Main Conference). arXiv:2601.07197
- Akter, S.*, **Shihab, I.F.***, and Sharma, A. “Detecting Proxy Gaming in RL and LLM Alignment via Evaluator Stress Tests.” **ACL 2026** (Findings). arXiv:2507.05619

- **Shihab, I.F.***, Akter, S.*, Sharma, A. “LLM-Guided Probabilistic Fusion for Label-Efficient Document Layout Analysis.” **CVPR 2026**. arXiv:2511.08903
- Akter, S.*, **Shihab, I.F.***, and Sharma, A. “Image Segmentation with Large Language Models: A Survey with Perspectives for Intelligent Transportation Systems.” **IEEE Trans. on Intelligent Transportation Systems**, 2026. arXiv:2506.14096
- **Shihab, I.F.***, Akter, S.*, Sharma, A. “Efficient unstructured pruning of mamba state-space models for resource-constrained environments.” **EMNLP 2025**. doi: 10.18653/v1/2025.emnlp-main.562
- **Shihab, I.F.***, Akter, S.*, Sharma, A. “Cache-Efficient Posterior Sampling for Reinforcement Learning with LLM-Derived Priors Across Discrete and Continuous Domains.” **EMNLP 2025**. doi: 10.18653/v1/2025.emnlp-main.560
- **Shihab, I.F.**, Akter, S., Sharma, A. “HMAE: Self-Supervised Few-Shot Learning for Quantum Spin Systems.” **ECAI 2025**. doi: 10.3233/FAIA251031
- Sivaraman, A.L.*, Adu-Gyamfi, K.*, **Shihab, I.F.***, and Sharma, A. “ClearVision: Leveraging CycleGAN and SigLIP-2 for Robust All-Weather Classification in Traffic Camera Imagery.” **IEEE ITSC 2025**. doi: 10.1109/ITSC60802.2025.11423032
- Bhagat, S.R.*, **Shihab, I.F.***, and Sharma, A. “Accuracy Is Not Agreement: Expert-Aligned Evaluation of Crash Narrative Classification Models.” **IEEE ITSC 2025**. doi: 10.1109/ITSC60802.2025.11423656
- Ahmed, S., **Shihab, I.F.**, and Khokhar, A. “Quantum-driven Zero Trust Architecture with Dynamic Anomaly Detection in 7G Technology.” **Measurement: Digitalization**, 100005, 2025. doi: 10.1016/j.measdig.2025.100005
- **Shihab, I.F.**, Alvee, B.I., Sharma, A. “Leveraging Video-LLMs for Crash Detection and Narrative Generation.” **TRC-30, 2024**. [PDF]
- Rahman, M.S., **Shihab, I.F.**, Chu, L., and Sharma, A. “Deeplocalization: Using Change Point Detection for Temporal Action Localization.” **IEEE/CVF CVPR Workshops**, pp. 7252–7260, 2024. doi: 10.1109/CVPRW63382.2024.00721
- Ferdous, S., **Shihab, I.F.**, Chowdhury, R., and Reuel, N.F. “Reinforcement Learning-Guided Control Strategies for CAR T-Cell Activation.” **Biotechnology and Bioengineering**, 121(9), 2868-2880, 2024. doi: 10.1002/bit.28753
- **Shihab, I.F.**, Bhagat, S.R., and Sharma, A. “Robust and Precise Sidewalk Detection with Ensemble Learning.” **IEEE ITSC**, pp. 5092-5099, 2023. doi: 10.1109/ITSC57777.2023.10422138
- Ferdous, S., **Shihab, I.F.**, and Reuel, N.F. “Effects of Sequence Features on Machine-Learned Enzyme Classification Fidelity.” **Biochemical Engineering Journal**, 187, 108612, 2022. doi: 10.1016/j.bej.2022.108612
- **Shihab, I.F.**, Akter, S., and Sharma, A. “Calibrated Adaptation: Bayesian Stiefel Manifold Priors for Reliable Parameter-Efficient Fine-Tuning.” arXiv:2602.17809
- **Shihab, I.F.**, Akter, S., and Sharma, A. “Anytime-Valid Answer Sufficiency Certificates for LLM Generation via Sequential Information Lift.” arXiv:2510.06478

Under Review at EMNLP 2026

- Akter, S.*, **Shihab, I.F.***, and Sharma, A. “Selective Risk Certification for LLM Outputs via Information-Lift Statistics: PAC-Bayes, Robustness, and Skeleton Design.” arXiv:2509.12527
- **Shihab, I.F.**, Akter, S., and Sharma, A. “Differentiable Range-Partition Entropy for Instance-Optimal Geometric Algorithms.” arXiv:2509.03733

Under Review at BMVC 2026

- **Shihab, I.F.**, and Sharma, A. “Enhancing Traffic Incident Response through Sub-Second Temporal Localization with HybridMamba.” arXiv:2504.03235
- **Shihab, I.F.**, Akter, S., and Sharma, A. “Temporal Zoom Networks: Distance Regression and Continuous Depth for Efficient Action Localization.” arXiv:2511.03943

Under review at NeurIPS 2026

- **Shihab, I.F.**, Akter, S., and Sharma, A. “Learning to Forget Attention: Memory Consolidation for Adaptive Compute Reduction (CRAM).” arXiv:2602.12204
- **Shihab, I.F.**, Akter, S., and Sharma, A. “On the Sparsifiability of Correlation Clustering: Approximation Guarantees under Edge Sampling.” arXiv:2602.13684
- **Shihab, I.F.**, Akter, S., and Sharma, A. “Grassmannian Mixture-of-Experts: Concentration-Controlled Routing on Subspace Manifolds.” arXiv:2602.17798

Under Review at IEEE Trans. on Intelligent Transportation Systems

- **Shihab, I.F.**, Akter, S., and Sharma, A. “Calibrated and Resource-Aware Super-Resolution for Reliable Driver Behavior Analysis.” arXiv:2509.23535

Preprints

- Akter, S., **Shihab, I.F.**, and Sharma, A. “Large Language Models for Crash Detection in Video: A Survey of Methods, Datasets, and Challenges.” arXiv:2507.02074

GRANTS & FUNDING

- **Iowa DOT Transportation Safety Projects:** Contributed to \$500K+ in funded research projects developing AI-based navigation and crash detection systems (2022-2025). Role: Graduate Research Assistant.
- **Future Funding Plans:** Preparing NSF CRII proposal on ”LLM-Guided Reinforcement Learning for Autonomous Vehicle Safety” (Target: \$175K, Submission: Spring 2026).

AWARDS & HONORS

- **Top-Tier Conference Acceptances:** 24+ papers accepted including premier venues like ICML '26 (4) ACL '26 (4: 3 Main + 1 Findings), CVPR '26 (1), CVPR Workshops '24 (1), EMNLP '25 (2), ECAI '25 (1), IEEE ITSC ('23, '25 $\times 2$, 26 $\times 2$), and IEEE Trans. on Intelligent Transportation Systems '26 (1).
- **Research Impact:** H-index: 7, Total Citations: 170+, with publications spanning AI, Transportation, Quantum Computing, and Bioengineering.
- **Graduate Assistantship:** Fully-funded Ph.D. and M.S. assistantships at Iowa State University (2020-2025).

ACADEMIC SERVICE

- **Peer Review:** Program Committee member and reviewer for: Neurips ('26), IEEE/CVF CVPR ('24, '25, '26), ICCV '25, IEEE ITSC ('23-'26), ACM Multimedia ('25, '26), WACV ('26, '27), ICLR ('26), ICML ('26), ACL Industry Track ('26), EMNLP ('26). Total reviews: 60+.
- **Departmental Service:** Graduate Student Representative, Iowa State University Computer Science Committee (2022-2023); Member, University of Vermont Graduate Senate Education Committee (2019-2020).
- **Community Outreach:** Judge, 2025 State Science & Technology Fair of Iowa (SSTFI).

SELECTED INDUSTRY EXPERIENCE

- **Applied Scientist II, Amazon, Sunnyvale, CA** *January 2026 – Present*
 - **Traffic Intelligence & Analytics:** Developing applied machine learning and statistical models for traffic engineering problems like anomaly and bot detection, including traffic flow analysis, congestion modeling, and large-scale transportation data understanding.
 - **Data-Driven Optimization:** Designing data-driven methods to support traffic optimization, monitoring, and decision-making under real-world operational constraints.
 - **Scalable Modeling Pipelines:** Building scalable data pipelines and evaluation frameworks for experimentation, model validation, and deployment in production environments.
 - **Cross-Functional Collaboration:** Collaborating with traffic engineers, software engineers, and product stakeholders to translate domain requirements into robust applied ML solutions.
- **Applied Scientist Intern, Amazon, Seattle, WA** *June – August 2025*
 - **Knowledge Graph Research:** Developed novel Knowledge Graph Embedding model and negative sampling technique for sparse graphs. Built recommender system for automatic model architecture selection.
- **Data Scientist, SoilSerdem, Ames, IA** *January – December 2024*
 - **Precision Agriculture ML:** Engineered precision soil mapping engine boosting accuracy by 35% for 10+ farms. Optimized cloud architecture reducing costs by 20%.

TECHNICAL EXPERTISE

- **AI & Machine Learning:** Deep Learning (PyTorch, Keras), Reinforcement Learning (Ray, RLlib), Computer Vision (OpenCV), Large Language Models (LangChain, Ollama), Generative AI
- **Specialized Research:** Traffic Simulation (SUMO, CARLA, Isaac Gym), Quantum Neural Networks (torchquantum, Qiskit, PennyLane), Knowledge Graph Embedding
- **Programming & Tools:** Python, C++, Java, R, MATLAB; AWS, Git, Docker; Statistical Analysis, Experimental Design